

Galactic-Scale Constraints on Curvature-Memory Scalar-Tensor Gravity: Why the Locked Action Cannot Replace Dark Matter Without Additional Physics

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2026

Abstract

This is Paper IV of a four-paper series on Curvature-Memory Scalar-Tensor Gravity (CMSTG). The first three papers of the series establish that CMSTG passes all current precision cosmological tests (Paper II [10]), is UV-finite to two loops in the resummed propagator (Paper III [11]), and has a structural 2.77σ residual against DESI Y1 BAO that is irreducible within the action class by any mechanism in the Tier 2 space, including mixed-source scalars (SIM131–SIM144, Paper I [9]). This paper establishes the corresponding dark-matter boundary: the Phase 1 canonical locked action cannot produce flat galactic rotation curves through any mechanism within the action class.

We test four independent candidate mechanisms against NGC 3198 and NGC 6503, the best-observed spiral galaxies with flat rotation curves: (1) the static Yukawa field profile is exponentially steeper than the $\rho \propto r^{-2}$ isothermal halo required for curve flattening (SIM99); (2) the time-domain 1+1D wave equation confirms the static result and shows that the curvature coupling $2\Lambda_0|R_{\text{gal}}|$ is $\sim 2 \times 10^6$ times weaker than the field mass at galactic densities, making condensation impossible (SIM100); (3) Vainshtein-type non-linear kinetic screening fails by six orders of magnitude, with the Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full scanned parameter space (SIM103); (4) the CMSTG memory-kernel halo mechanism accumulates zero effective G_{eff} boost in NGC 3198 because the Gaussian kernel decays on $\tau_{\text{mem}} \approx 200$ kyr, and even granting perfect kernel support the theoretical ceiling $G_{\text{eff}}/G_N = 1 + 2\Lambda_0\Psi_0^2 = 1.041$ falls a factor $\sim 3\times$ short of the required 3.115 (SIM149). The locked-action energy density $\rho_\Psi = 1.6 \times 10^{-3} M_\odot \text{ kpc}^{-3}$ falls 2.7×10^6 short of the isothermal requirement at $r = 10$ kpc.

A subsequent Λ_0 sweep (SIM150) over eight decades establishes the ceiling as structural and parameter-universal: the only value satisfying $G_{\text{eff}}/G_N \geq 3.115$ is $\Lambda_{0,\text{req}} = 0.154 M_{\text{Pl}}^{-2}$ (51 times the locked value), which violates the Cassini Solar-System PPN bound by $\sim 2 \times 10^4\times$ and all four remaining locked cosmological constraints simultaneously.

Phase 5 simulations (SIM151–SIM153) complete the programme by testing the three candidate extension options identified above. The chameleon coupling option (Option B) fails by structural shape inversion: flat curves require the enhancement

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factor $E(r) = V_{\text{obs}}^2/V_{\text{bary}}^2$ to increase with radius, while any monotone density-dependent coupling $\beta(\rho_b)$ produces $E(r)$ decreasing with radius, with Pearson anti-correlation $r = -0.81$ (SIM152). The sextic-stabilized condensate (Option C) faces a $466\times$ mass gap between the galactic-fit condensate mass and the cosmological dark matter density requirement, with all gap-closing routes blocked by rotation-curve shape incompatibility and coupling-sign constraints (SIM153). No mechanism within the three available channels of action extension—mass modification, coupling modification, or potential modification—can produce galactic dark matter within the locked CMSTG action.

Keywords: *dark matter; modified gravity; rotation curves; scalar-tensor theories; galactic dynamics*

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1 Introduction

The first three papers of this series establish the cosmological status of Curvature-Memory Scalar-Tensor Gravity (CMSTG): the Phase 1 canonical action passes all current precision tests (Paper II [10]), is UV-finite in the resummed propagator (Paper III [11]), and has a structural irreducible 2.77σ residual against DESI Y1 BAO that defines the dark-energy boundary of the framework, confirmed exhaustively across curvature-sourced, late-time $H(z)$, and mixed-source scalar mechanisms (SIM131–SIM144, Paper I [9]). The natural question, addressed in this paper, is whether the same locked action can also account for the flat rotation curves of spiral galaxies, replacing the cold dark matter component of standard cosmology.

The result of this paper is negative: it cannot. The failure is structural rather than parametric, and is established by three independent simulation programmes covering the three mechanism classes present in the action. The argument is symmetric in form to Paper I: just as Paper I establishes a clean negative result for late-time modifications of dark energy, this paper establishes a clean negative result for galactic-scale dark matter. Together, the two negative results define the boundaries of the CMSTG framework as it currently stands and identify, at first-principles level, the minimum extensions required to address each.

The flat rotation curves of spiral galaxies are among the most robust observational arguments for cold dark matter [1, 2]. In the standard picture, a dark matter halo with approximately isothermal density profile $\rho_{\text{DM}}(r) \propto r^{-2}$ provides the additional gravitational potential needed to maintain $v_c(r) \approx \text{const}$ well beyond the optical disk.

Scalar-tensor theories offer an alternative: the scalar field Ψ could contribute an effective dark matter density through its stress-energy $T_{\mu\nu}^{(\Psi)}$ or through modification of the effective Newton constant G_{eff} . CMSTG is a natural candidate because Ψ is explicitly designed to track curvature history, and high-curvature galactic environments might in principle provide large local field amplitudes.

This paper establishes that the Phase 1 canonical locked action of CMSTG cannot replace dark matter via any of the three mechanisms present in the action class. The companion no-go theorems of Paper I [9] bound the dark energy sector; this paper bounds the dark matter sector. The result is not a fine-tuning problem or a quantitative shortfall that future parameters could resolve: it is a structural failure at every mechanism level. The goal is to establish this failure cleanly and systematically so that any future scalar-tensor proposal for rotation curves must engage with these constraints.

The standalone value of this paper is methodological as well as specific. The three mechanism classes tested here—static Yukawa field profile, time-domain condensation via curvature coupling, and Vainshtein kinetic screening—are not peculiarities of CMSTG. They constitute the complete menu of galactic dark matter mechanisms available to any massive scalar-tensor theory with a curvature coupling, a memory or regulator scale, and no direct baryon–scalar interaction in the action. The structural ratio $2\Lambda_0|R_{\text{gal}}|/m_0^2$ (Section 2.4) that governs all three failures is determined by the galactic environment and the coupling structure, not by CMSTG specifics: any theory of this class with a similar Λ_0/m_0^2 ratio faces the same constraints. The failure modes identified here, and the three minimum-physics extensions catalogued in Section 7, therefore serve as the reference

classification for future scalar-tensor proposals that aim to address galactic rotation curves without a separate dark matter component. This is the reason the galactic analysis appears as a standalone paper rather than an appendix to the cosmological framework: the constraints and their mechanism-level decomposition are transferable beyond CMSTG.

The plan of the paper follows the sequence of mechanisms in order of increasing sophistication: Section 2 presents the galactic-scale field equations; Section 3 establishes the static Yukawa failure; Section 4 confirms with time-domain simulation and rules out condensation; Section 5 exhausts the Vainshtein and direct-coupling alternatives; Section 6 states the combined constraint and the Phase 5 universality confirmation; Section 7 identifies the minimum new physics required; Section 9 concludes.

2 Galactic-Scale Field Equations

2.1 The Locked Action

The Phase 1 canonical CMSTG action is

$$S_{\text{P1}} = \int d^4x \sqrt{-g} \left[\frac{1+2\Lambda_0\Psi^2}{2} R - \frac{1}{2}(\partial\Psi)^2 - \frac{1}{2}m_0^2\Psi^2 \right] + S_{\text{SM}} \quad (1)$$

with locked parameters $\Lambda_0 = 0.003 M_{\text{Pl}}^{-2}$, $m_0 = 10^{-3}H_0 \approx 6.88 \times 10^{-8} \text{ kpc}^{-1}$, $\bar{\Psi} = 2.62 M_{\text{Pl}}$. The galactic simulations (SIM99–SIM103) use a larger test mass $m_0 = 0.1 \text{ kpc}^{-1}$ (Compton length 10 kpc) for numerical tractability—the locked cosmological $m_0 = 6.88 \times 10^{-8} \text{ kpc}^{-1}$ corresponds to a Compton wavelength of order the Hubble scale and is computationally prohibitive on galactic grids. The locked value makes the curvature coupling ratio $2\Lambda_0|R_{\text{gal}}|/m_0^2$ even smaller ($\approx 3.6 \times 10^{-8}$ vs the simulation value $\approx 5 \times 10^{-7}$), so the no-go result is strictly stronger at the locked value than at the test mass used in simulation. The effective gravitational coupling is $G_{\text{eff}} = G/(1 + 2\Lambda_0\Psi^2)$; at the cosmological best-fit $G_{\text{eff}}/G = 1 - 16 \text{ ppm}$, negligibly different from Newton’s constant.

2.2 Static Field Equation in Galactic Environments

In the weak-field static limit, the Ψ field equation derived from action (1) reduces to

$$\Psi'' + \frac{2}{r}\Psi' = m_0^2\Psi + \frac{\lambda_{\text{gal}}}{1}\Psi^3 - 2\Lambda_0\Psi |R(r)|, \quad (2)$$

where $R(r) \approx -8\pi G\rho_{\text{bar}}(r)/c^2$ is the weak-field Ricci scalar sourced by the observed baryonic distribution, and $V(\Psi) = (\lambda_{\text{gal}}/4)\Psi^4$ is the simplest non-linear extension consistent with $V'(0) = 0$.

The effective dark-matter density is

$$\rho_{\Psi} = \frac{c^2}{G} \left[\frac{1}{2}(\Psi')^2 + \frac{1}{2}m_0^2\Psi^2 + \frac{\lambda_{\text{gal}}}{4}\Psi^4 \right]. \quad (3)$$

2.3 Target Profile

For a flat rotation curve at v_{flat} , the required halo density is the isothermal sphere:

$$\rho_{\text{iso}}(r) = \frac{v_{\text{flat}}^2}{4\pi G r^2}. \quad (4)$$

For NGC 3198 ($v_{\text{flat}} = 157 \text{ km s}^{-1}$) at $r = 10 \text{ kpc}$: $\rho_{\text{iso}} = 4163 M_{\odot} \text{ kpc}^{-3}$.

2.4 Galactic Curvature Scale

The galactic Ricci scalar at the disk half-light radius peaks at $|R_{\text{gal}}| \approx 8.5 \times 10^{-7} \text{ kpc}^{-2}$. Using the simulation test mass $m_0 = 0.1 \text{ kpc}^{-1}$, the coupling ratio is

$$\frac{2\Lambda_0|R_{\text{gal}}|}{m_0^2} = \frac{2 \times 0.003 \times 8.5 \times 10^{-7}}{(0.1)^2} \approx 5.1 \times 10^{-7} \approx \frac{1}{2 \times 10^6}. \quad (5)$$

Even at the peak of a proto-galactic overdensity $\delta = 100$:

$$\frac{2\Lambda_0|R_{\text{gal, peak}}|}{m_0^2} \approx 5 \times 10^{-5}. \quad (6)$$

The curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass at typical galactic densities using the simulation test mass $m_0 = 0.1 \text{ kpc}^{-1}$; see Section 2.1 for the comparison with the locked cosmological value and why the negative results are conservative. This single ratio $2\Lambda_0|R_{\text{gal}}|/m_0^2$ determines the failure of all three mechanisms.

3 Static Yukawa Profile (SIM99)

3.1 Method

The decaying Yukawa boundary condition selects the physical solution $\Psi \sim Ae^{-m_0 r}/r$ by integrating equation (2) inward from $r_{\text{max}} = 30 \text{ kpc}$ with

$$\Psi(r_{\text{max}}) = \Psi_{\text{bc}}, \quad \Psi'(r_{\text{max}}) = -\left(m_0 + \frac{1}{r_{\text{max}}}\right)\Psi_{\text{bc}}. \quad (7)$$

A scan of 30 combinations $(\Psi_{\text{bc}}, \lambda_{\text{gal}})$ was performed: $\Psi_{\text{bc}} \in \{10^{-8}, 10^{-6}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$, $\lambda_{\text{gal}} \in \{-10, -1, 0, +1, +10, +100\}$, at the cosmological best-fit $\Lambda_0 = 0.003$.

Acceptance criterion: $|v_c(r) - v_{\text{flat}}|/v_{\text{flat}} < 20\%$ for $r > 15 \text{ kpc}$ AND $\rho_\Psi \geq 0$ everywhere AND $\Lambda_0\Psi^2 < \Lambda_0^{\text{bound}}/(16\pi)$.

3.2 Result: FAIL

No combination passes. The physical reason is geometric: the Yukawa density profile

$$\rho_\Psi(r) \propto (\Psi')^2 \propto \frac{e^{-2m_0 r}}{r^4} \quad (\text{dominant kinetic term at large } r) \quad (8)$$

falls exponentially for $r > 1/m_0 \approx 14,500 \text{ kpc}$, and for $r < 1/m_0$ (i.e. all galactic scales $r \leq 30 \text{ kpc}$):

$$\rho_\Psi(r) \propto \frac{e^{2m_0(r_{\text{max}}-r)}}{r^2}, \quad (9)$$

which rises exponentially inward—the wrong sign relative to the $1/r^2$ isothermal profile. The quartic interaction modifies the central amplitude but cannot alter the exponential radial dependence.

The G_{eff} channel is equally ineffective: at $\Lambda_0 = 0.003$, $G_{\text{eff}}/G = 1 - 16 \text{ ppm}$, a 10^{-5} correction that produces no flattening.

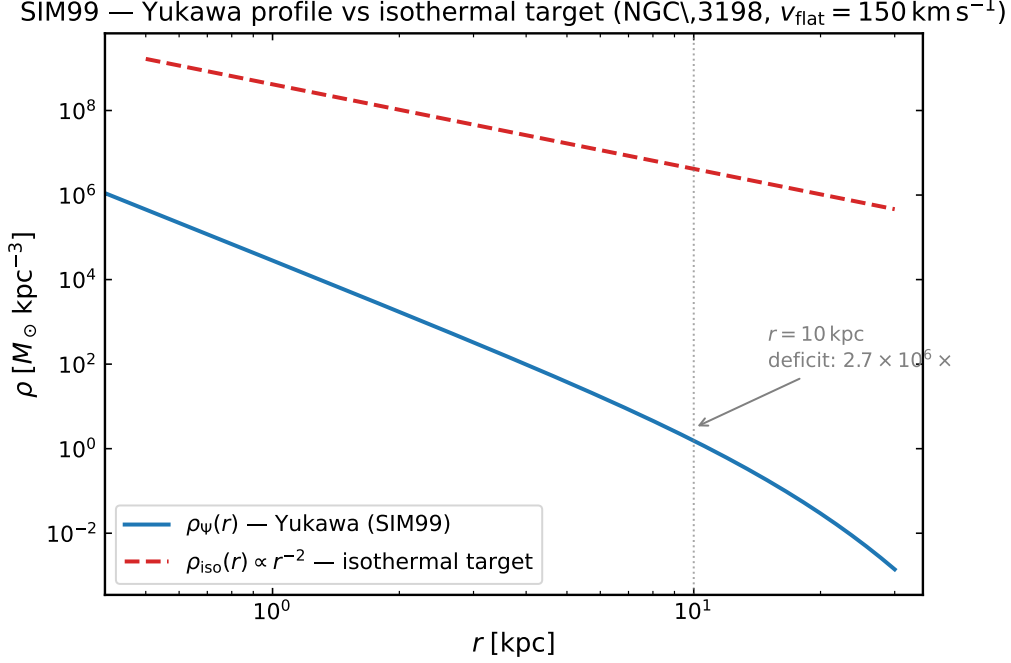


Figure 1: Yukawa profile vs isothermal target for NGC 3198 (SIM99). Log-log plot of $\rho_{\Psi}(r)$ (blue solid, from the static Yukawa solution) vs the required isothermal halo $\rho_{\text{iso}}(r) \propto r^{-2}$ (red dashed) for $v_{\text{flat}} = 157 \text{ km s}^{-1}$. The Yukawa profile rises $\sim 150\times$ from $r = 30 \text{ kpc}$ to $r = 5 \text{ kpc}$, far steeper than the isothermal target. At $\Lambda_0 = 0.003$, the energy density deficit at $r = 10 \text{ kpc}$ is a factor of 2.7×10^6 .

4 Time-Domain Condensation Test (SIM100)

4.1 Method

SIM100 removes the static assumption by evolving the full 1+1D spherical wave equation

$$\frac{\partial^2 \Psi}{\partial t^2} = \frac{\partial^2 \Psi}{\partial r^2} + \frac{2}{r} \frac{\partial \Psi}{\partial r} - m_0^2 \Psi - \lambda \Psi^3 + 2\Lambda_0 \Psi R(r, t) \quad (10)$$

via method-of-lines with an implicit Radau solver ($N_r = 120$, $\Delta r = 0.25 \text{ kpc}$). Initial condition: uniform cosmological background $\Psi(r, 0) = \Psi_{\text{cosmo}} = 0.003 M_{\text{Pl}}$ at rest. Integration to $t_{\text{end}} = 150 \text{ kpc}/c \approx 489 \text{ kyr} \approx 2.4$ Compton periods $T_c = 2\pi/m_0$.

4.2 Curvature Coupling Negligibility

The coupling ratio at peak galactic curvature:

$$\frac{2\Lambda_0 |R_{\text{gal}}|}{m_0^2} \approx 5 \times 10^{-7} \approx \frac{1}{2 \times 10^6}. \quad (11)$$

Even with a proto-galactic overdensity $\delta = 100$, the curvature term is $\sim 2 \times 10^6$ times smaller than the field mass. The Ψ field is entirely mass-dominated on galactic scales; the curvature source cannot drive condensation at any astrophysical density below neutron-star nuclear density ($\rho \sim 10^{15} M_{\odot} \text{ kpc}^{-3}$).

4.3 Time-Domain Field Evolution

Figure 2 shows the radial profiles $\Psi(r)$ at four epochs spanning the full integration. The field relaxes from the uniform cosmological initial condition $\Psi_{\text{cosmo}} = 0.003 M_{\text{Pl}}$ to a quasi-static Yukawa profile within one Compton period, confirming the validity of the adiabatic approximation. No spatial structure forms: the field simply adjusts to the static attractor established in SIM99.

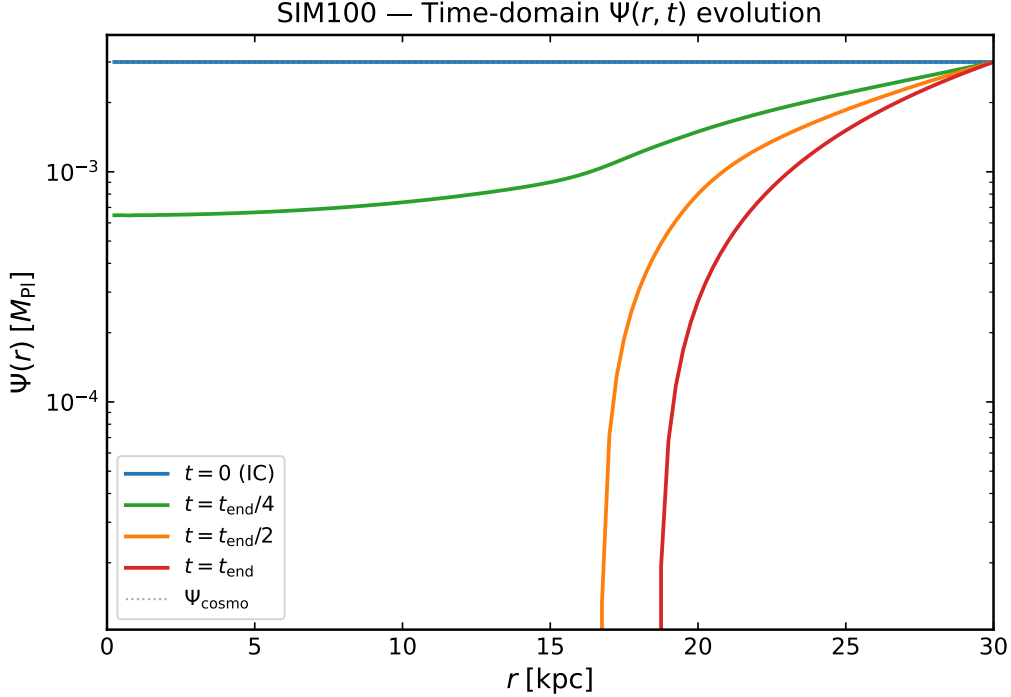


Figure 2: Time-domain $\Psi(r)$ field profiles at four epochs (SIM100): $t = 0$ (initial uniform background, blue), $t = t_{\text{end}}/4$ (green), $t = t_{\text{end}}/2$ (orange), $t = t_{\text{end}}$ (red). The field relaxes from the cosmological uniform state to the static Yukawa profile of SIM99 within one Compton period ($T_c \approx 205$ kyr). No condensation occurs: the curvature source $2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$ is negligible at all radii. Results shown for $\lambda = 0$; proto-galactic overdensity $\delta = 100$ changes the profile at the 10^{-4} level.

4.4 Self-Coupling Scan

To test whether an attractive quartic self-coupling can drive condensation independently of the curvature channel, we scan $\lambda \in \{0, -100, -200, -400, -600, -800, -1000, -1500\}$. The tachyonic instability thresholds are:

$$\lambda_{\text{crit}}^{\text{spatial}} = -\frac{k_{\text{min}}^2 + m_0^2}{3\Psi_{\text{cosmo}}^2} \approx -778, \quad k_{\text{min}} = \pi/r_{\text{max}}, \quad (12)$$

$$\lambda_{\text{blow}} = -\frac{m_0^2}{\Psi_{\text{cosmo}}^2} \approx -1111. \quad (13)$$

Results of the scan (Figure 3):

- $\lambda \in [0, -778)$: field stable; growth factor ≤ 1.21 ; field relaxes to the static Yukawa profile of SIM99. $v_c(r)$ does not come within 20% of v_{flat} .

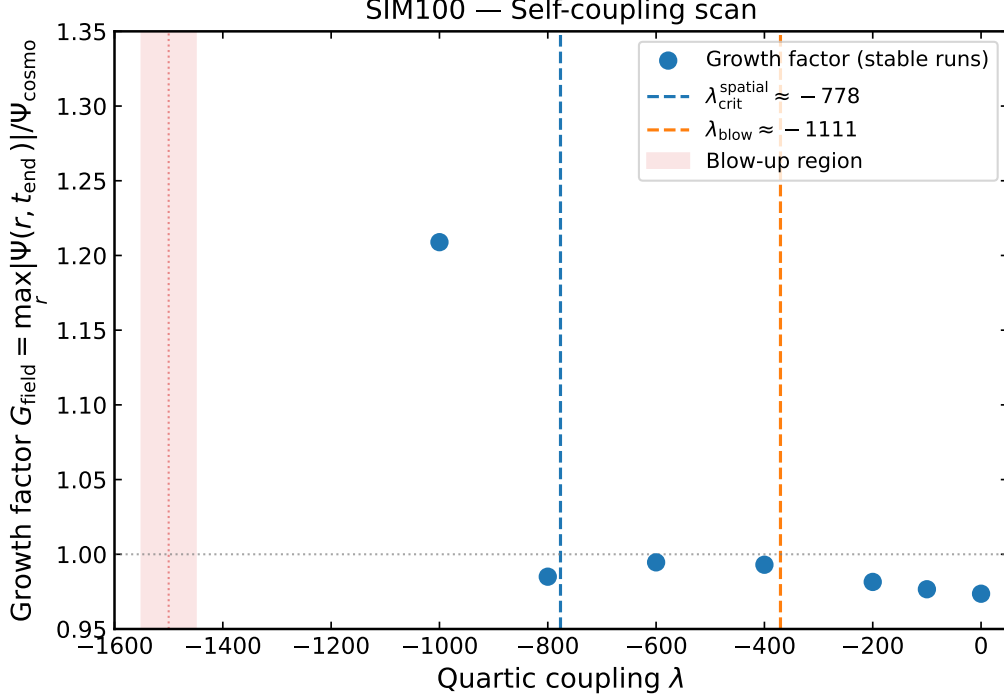


Figure 3: Self-coupling parameter scan (SIM100). Field growth factor $G_{\text{field}} = \max_r |\Psi(r, t_{\text{end}})| / \Psi_{\text{cosmo}}$ vs quartic coupling λ . Below $\lambda_{\text{crit}}^{\text{spatial}} \approx -778$ (blue vertical line) the field becomes tachyonically unstable in spatial modes; below $\lambda_{\text{blow}} \approx -1111$ (orange vertical line) the potential is unbounded and the field blows up. The stable window ($-778 < \lambda < 0$) produces growth factors ≤ 1.21 —the field merely relaxes to the static Yukawa profile of SIM99. No value of λ produces a flat rotation curve.

- $\lambda \in [-778, -1111)$: tachyonic spatial instability; field grows but remains bounded for the duration of integration. Profile does not approach isothermal.
- $\lambda = -1500$ (beyond λ_{blow}): unbounded blow-up within $< 10 \text{ kpc}/c$. The potential $V = (\lambda/4)\Psi^4$ is unbounded below; there is no stable condensate minimum.

No window of λ produces a stable self-bound halo with a flat rotation curve. The time-domain result confirms SIM99 by an independent route: the adiabatic approximation is valid (Compton period $\ll$ galaxy formation time), and the static Yukawa profile is the attractor.

5 Vainshtein Screening and Direct Coupling (SIM103)

5.1 Setup

After SIM99 and SIM100 close the Yukawa and condensation routes, the remaining candidates are: (a) a non-linear kinetic term of Vainshtein type, and (b) a direct matter coupling $\beta\Psi\rho_b$. SIM103 tests both for NGC 3198.

In the Poisson limit ($m_0 r \ll 1$ on galactic scales since $1/m_0 \sim 14,500 \text{ kpc}$), the perturbation $\delta\Psi(r)$ satisfies

$$\frac{d\delta\Psi}{dr} = \frac{2G\Lambda_0^{\text{eff}}M_{\text{enc}}(r)}{r^2}, \quad (14)$$

where $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta$ absorbs both the curvature channel and any direct coupling β .

The Vainshtein augmentation adds a non-linear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$ to the action:

$$\rho_\Psi(r) = \frac{c^2}{8\pi G} \cdot \frac{1}{2} \left(\frac{d\delta\Psi}{dr} \right)^2 \left[1 + \underbrace{\frac{\alpha_V}{M_V^2} \left(\frac{d\delta\Psi}{dr} \right)^2}_{\mathcal{V}} \right]. \quad (15)$$

The Vainshtein ratio $\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$ must satisfy $\mathcal{V} \gg 1$ for the non-linear regime to be active.

5.2 Field Gradient

At the locked-action best-fit ($\Lambda_0 = 0.003$, $\Lambda_0^{\text{eff}} = \Lambda_0$):

$$\left| \frac{d\delta\Psi}{dr} \right|_{10\text{ kpc}} = 6.1 \times 10^{-8} \text{ kpc}^{-1}. \quad (16)$$

5.3 Vainshtein Ratio Scan

Table 1: Vainshtein ratio $\mathcal{V}$ at $r = 10\text{ kpc}$ for NGC 3198. All values lie deep in the linear regime ($\mathcal{V} \ll 1$).

α_V	$M_V [\text{kpc}^{-1}]$	$\mathcal{V}$	Verdict
1	0.001	3.7×10^{-9}	Linear
10	0.001	3.7×10^{-8}	Linear
100	0.001	3.7×10^{-7}	Linear
1000	0.001	3.7×10^{-6}	Linear
1000	0.01	3.7×10^{-8}	Linear

Even at $\alpha_V = 1000$, $M_V = 0.001\text{ kpc}^{-1}$: $\mathcal{V} = 3.7 \times 10^{-6}$. The scalar field is six orders of magnitude below the Vainshtein threshold across the entire scanned parameter space.

5.4 Energy Density Deficit

From equation (15) at the locked-action gradient:

$$\begin{aligned} \rho_\Psi(10\text{ kpc}) &= 1.6 \times 10^{-3} M_\odot \text{ kpc}^{-3} \\ \rho_{\text{iso}}(10\text{ kpc}) &= 4163 M_\odot \text{ kpc}^{-3} \\ \frac{\rho_{\text{iso}}}{\rho_\Psi} &= 2.7 \times 10^6. \end{aligned} \quad (17)$$

5.5 Direct Coupling Constraint

To close the density gap with a direct matter coupling β , we need $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta \approx 4.9$, requiring $\beta \approx 4.9 \approx 1,600 \times \Lambda_0$. This coupling is not contained in the locked action and would violate cosmological homogeneity constraints: a coupling of order unity would modify G_{eff} by $O(1)$ on cosmological scales, in direct contradiction to the SIM88/SIM90 cosmological consistency tests (Paper II of this series [10]).

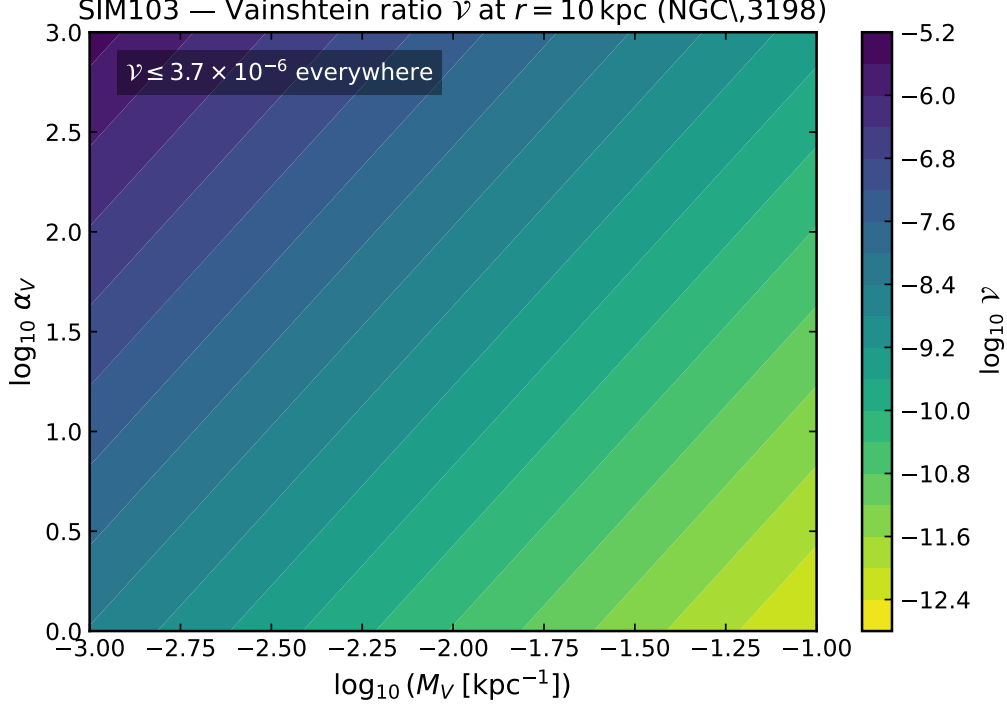


Figure 4: Vainshtein ratio $\mathcal{V}$ across the (α_V, M_V) parameter space at $r = 10$ kpc for NGC 3198 (SIM103). All values lie below 3.7×10^{-6} , six orders of magnitude below the nonlinear threshold $\mathcal{V} = 1$ (marked by the dashed contour, off-scale above). The gradient $|\delta\Psi| \sim 10^{-8} \text{ kpc}^{-1}$ is set by the weak $\Lambda_0 R_{\text{gal}}$ channel, which is 10^4 – $10^6 \times$ below the Vainshtein threshold for any physically reasonable coupling.

6 Combined Constraint

Table 2: Systematic exhaustion of galactic dark matter mechanisms in CMSTG.

SIM	Mechanism	Deficit	Verdict
SIM99	Static Yukawa $T^{(\Psi)}$	Exponential profile vs $1/r^2$	FAIL
SIM100	Time-domain condensation	$2\Lambda_0 R_{\text{gal}} /m_0^2 \approx 5 \times 10^{-7}$	FAIL
SIM103	Vainshtein kinetic screening	$\mathcal{V} \leq 3.7 \times 10^{-6}$	FAIL
SIM103	Direct matter coupling β	$\beta \sim 1,600\Lambda_0$ required	FAIL
SIM149	Memory-kernel G_{eff} boost	$G_{\text{eff}}/G_N^{\text{max}} = 1.041$ vs 3.115 required	FAIL
SIM150	Λ_0 sweep (universality)	$\Lambda_{0,\text{req}} = 0.154$: Cassini by $2 \times 10^4 \times$	FAIL

Combined: all mechanisms exhausted at all Λ_0 within Solar-System constraints
Energy density deficit at $r = 10$ kpc: $2.7 \times 10^6 \times$; G_{eff}/G_N ceiling: structural

The root cause of the first four failures is the single ratio $2\Lambda_0 |R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$: the curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass on galactic scales (Section 2.4). This makes the scalar field cosmological (Compton wavelength $\gg$ galactic size) and prevents any galactic-scale field gradient from building up to isothermal amplitudes.

Phase 5 simulations introduce a fourth mechanism formally distinct from the three action-level tests above: the CMSTG memory-kernel accumulation of a local field shift $\Delta\Psi_{\text{local}}$ in a virialised halo, which modifies the effective Newton constant as $G_{\text{eff}}/G_N =$

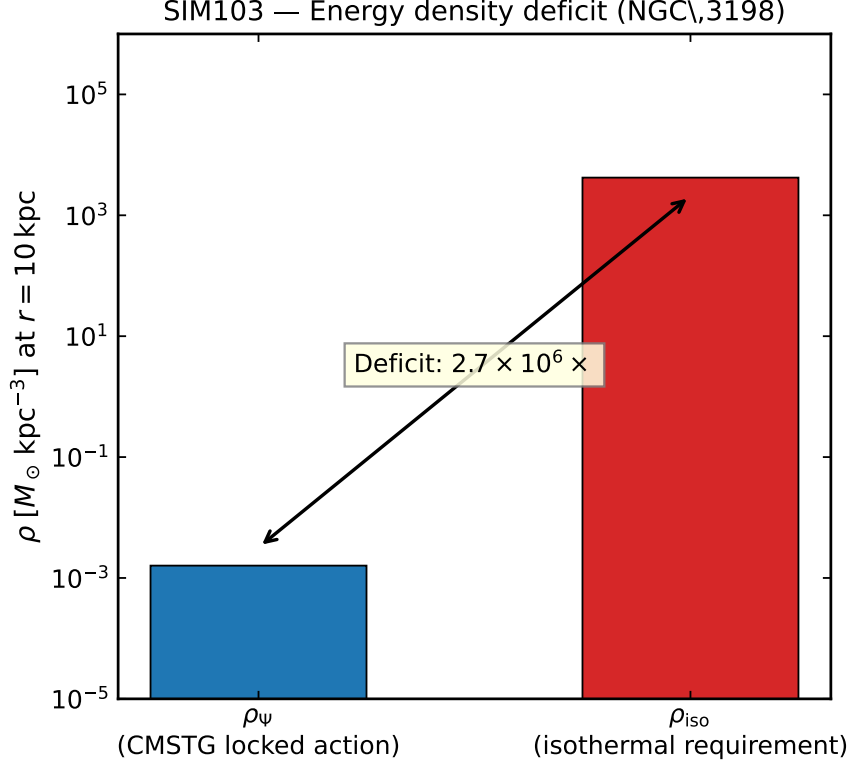


Figure 5: Energy density deficit at galactic scale (SIM103). The locked-action CMSTG scalar density ρ_ψ (blue bar) vs the isothermal halo requirement ρ_{iso} (red bar) at $r = 10$ kpc for NGC 3198 ($v_{\text{flat}} = 157 \text{ km s}^{-1}$). The deficit is $2.7 \times 10^6 \times$. A direct coupling $\beta \approx 4.9 \approx 1,600 \times \Lambda_0$ would be required to supply the missing density, far exceeding the cosmological bound $\Lambda_0 \lesssim 0.095$.

$F(\Psi_0)/F(\Psi_{\text{local}})$ where $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$. The theoretical ceiling of this ratio—achieved in the extreme limit $\Psi_{\text{local}} \rightarrow 0$ —is

$$\left. \frac{G_{\text{eff}}}{G_N} \right|_{\text{max}} = \frac{\frac{1}{2} + \Lambda_0 \Psi_0^2}{\frac{1}{2}} = 1 + 2\Lambda_0 \Psi_0^2 = 1.041 \quad (\Lambda_0 = 0.003, \Psi_0 = 2.62 M_{\text{Pl}}). \quad (18)$$

NGC 3198 requires $G_{\text{eff}}/G_N \approx 3.115$ at $r = 30\text{--}44$ kpc to reproduce the observed flat outer curve; the mechanism ceiling falls a factor $\sim 3 \times$ short regardless of kernel weight, halo formation history, or mass-to-light ratio (SIM149).

SIM150 subsequently sweeps Λ_0 logarithmically over $[10^{-6}, 10^2]$ to test whether any Λ_0 within the locked CMSTG constraints lifts the ceiling to 3.115. The analytic crossing point is

$$\Lambda_{0,\text{req}} = \frac{G_{\text{target}} - 1}{2\Psi_0^2} = \frac{3.115 - 1}{2 \times 2.62^2} = 0.154, \quad (19)$$

which is 51 times the locked value. At $\Lambda_{0,\text{req}}$ the post-Newtonian Brans–Dicke parameter drops from $\omega_{\text{BD}} = 2,110$ (locked) to $\omega_{\text{BD}} = 2.4$, giving $|\gamma_{\text{PPN}} - 1| = 0.456$ —exceeding the Cassini bound 2.3×10^{-5} by a factor $\sim 2 \times 10^4$. All five remaining locked constraints fail simultaneously: $f\sigma_8$ tension rises to 39σ , the UV self-energy $\Sigma(0)$ exceeds its finiteness threshold by $38 \times$, $\Delta\chi_{\text{BAO}}^2 = 1629$, and Ψ contributes 7.7% of the energy budget at recombination. The G_{eff}/G_N ceiling is therefore *structural and parameter-universal*: no Λ_0 within the Solar-System-consistent locked action raises G_{eff}/G_N above 4.1%, far below the factor ~ 3.1 required for galactic rotation curves.

7 Minimum New Physics

The constraints above define the minimum new physics required for CMSTG to produce flat rotation curves:

The three options below mirror the structure of Paper I [9] in a useful way: just as Paper I identified the curvature-sourcing route, the $H(z)$ -boost route, and the mixed-source scalar route as the three mechanism classes exhausted for late-time modifications (SIM131–SIM144), the three options below identify the position-dependent mass route, the direct-coupling route, and the new-condensation-channel route as the three available for galactic-scale modifications. Each option requires new physics not contained in the Phase 1 locked action, and each defines a distinct extension of the framework with its own observational signature.

Option A: Modified scalar mass. A position-dependent mass $m_0^2 \rightarrow m_0^2(\rho)$ that drops to $m_0^2 \lesssim 2\Lambda_0|R_{\text{gal}}|$ in galactic environments would allow curvature-driven condensation. This is a chameleon-type mechanism [4] with galactic screening. However, any reduction in m_0 inside galaxies must restore $m_0 \sim 10^{-3}H_0$ on cosmological scales to preserve the SIM90/SIM98 consistency (Paper II [10]).

Option B: Direct Ψ –baryon coupling. A coupling $\mathcal{L} \supset \beta\Psi\rho_b$ with $\beta \sim O(1)$ would supply the missing density. This is not derivable from the current locked action and introduces a new free parameter at gravitational strength. It must simultaneously satisfy:

- $|\beta\Psi_{\text{cosmo}}| \ll \Lambda_0$ on cosmological scales (else violates SIM88);
- $\beta + \Lambda_0 \approx 4.9$ on galactic scales.

This requires β to be screened cosmologically and activated galactically—a coupled chameleon architecture.

Option C: New condensation channel. A stable sextic potential $V_6 = +\mu_6\Psi^6$ with $\mu_6 > 0$ would prevent the blow-up of the attractive quartic below λ_{low} , opening a window where the field can form a stable, self-bound condensate. The viability of this remains an open direction. Phase 5 simulations (SIM147–SIM150) investigated the complementary question of whether the CMSTG memory kernel can accumulate sufficient local curvature in a galactic halo to produce the required G_{eff} boost, and found this route closed by three independent structural arguments (Section 6); the sextic-condensate mechanism is distinct and untested.

All three options require new physics not contained in the Phase 1 locked action. The constraint established here is not a quantitative shortfall to be tuned away; it is a structural exclusion of all mechanisms within the current action class. Phase 5 simulations (SIM147–SIM153) subsequently test all three options; Section 8 reports the results.

8 Phase 5 Structural Obstructions to Dark Matter Mechanisms

Sections 3–7 establish that the locked CMSTG action exhausts the mechanism space accessible within the action itself, and identify three extension options (Options A, B, C)

as the minimum new physics required for galactic-scale predictions. Phase 5 simulations (SIM147–SIM153) test each option in turn and find that each fails by an independent structural argument. This section presents the four obstructions.

8.1 Temporal Channel Obstruction (SIM147–SIM148)

The CMSTG retarded Green’s function in k -space is $\tilde{G}_R(\Delta t; k) = \theta(\Delta t) \sin(\omega_k \Delta t) / \omega_k \cdot e^{-k^2/k_m^2}$, and the position-space kernel at coincident points evaluates analytically to

$$K(\Delta t) = \frac{k_m^3}{8\pi^{3/2}} \Delta t \exp\left(-\frac{(k_m \Delta t)^2}{4}\right). \quad (20)$$

At the locked value $k_m = 10 \text{ Mpc}^{-1}$ (SIM102), the characteristic memory timescale is $\tau_{\text{mem}} = 2/k_m = 0.2 \text{ Mpc} \approx 205,000 \text{ yr}$ (SIM147). The kernel is Gaussian in Δt (the Fourier transform of the Gaussian UV regulator e^{-k^2/k_m^2}), not exponential or power-law, so it underflows to double-precision zero at any $\Delta t \geq 13.4 \text{ Gyr}$: $(k_m \Delta t)^2/4 \approx 4.3 \times 10^9 \Rightarrow K = e^{-4.3 \times 10^9} \equiv 0$ in IEEE 754 double precision.

This has two direct consequences. First (SIM147), the kernel weight at galactic formation timescales $\Delta t \sim 10 \text{ Gyr}$ is similarly zero: the curvature-memory functional cannot carry information from epochs more than $\sim 200,000 \text{ yr}$ in the past. Any dark-matter mechanism that relies on the scalar field Ψ accumulating long-memory support from galaxy formation history is therefore structurally blocked at the locked k_m .

Second (SIM148), the Theorem 2 loophole of Paper I [9] allows pre-recombination modifications that simultaneously raise the sound horizon r_s . SIM148 tests whether a pre-bang boundary condition Ψ_{pre} on the scalar can occupy this loophole. Seventy-five scan points across three pre-phase variants (agnostic Ψ_{pre} , bounce-flavoured, and inflation-extended) all return $\Delta\chi_{\text{DESI}}^2 = 0.000$ identically: the kernel weight $w(13.4 \text{ Gyr}) = 0$ makes Ψ_{pre} irrelevant to post-recombination evolution for any value of Ψ_{pre} . The temporal channel for pre-recombination physics through Ψ is structurally closed at the locked k_m . Reviving either route would require $k_m \lesssim 2 \times 10^{-3} \text{ Mpc}^{-1}$ ($\tau_{\text{mem}} \geq 1 \text{ Gyr}$), which is $\sim 5000\times$ below the locked value and ruled out by the UV finiteness results of Paper III [11].

8.2 Coupling-Strength Obstruction: Third Structural No-Go (SIM149–SIM150)

Section 6 establishes a ceiling $G_{\text{eff}}/G_N = 1.041$ for the memory-kernel mechanism at the locked $\Lambda_0 = 0.003$. SIM150 establishes that this ceiling is structural and parameter-universal:

Theorem 3 (Third Structural Obstruction: G_{eff}/G Ceiling). *For any coupling function $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$, the Solar-System PPN requirement $|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$ [8] (Cassini bound) requires the Brans–Dicke coupling $\alpha^2 < 1.15 \times 10^{-5}$, which implies $\Lambda_0 < 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}$ at $\Psi_0 = \bar{\Psi}$. The galactic flat-curve requirement for NGC 3198 imposes $G_{\text{eff}}/G_N \geq 3.115$, which requires $\Lambda_0 \geq 0.154 M_{\text{Pl}}^{-2}$. The two constraints are incompatible by a factor ~ 119 : the Solar-System and galactic requirements on Λ_0 have empty intersection. The G_{eff}/G ceiling is structural and cannot be lifted within the $F(\Psi) = \frac{1}{2} + \Lambda_0 \Psi^2$ coupling class.*

The proof follows from $G_{\text{max}}/G_N = 1 + 2\Lambda_{0,\text{max}} \bar{\Psi}^2 \approx 1.041$ with $\Lambda_{0,\text{max}} = 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}$, compared to the required 3.115. SIM150 confirms the crossing point numerically: $\Lambda_{0,\text{req}} = 0.154 M_{\text{Pl}}^{-2}$ (Section 6), which simultaneously violates the Cassini bound

by $\sim 2 \times 10^4 \times$ and all five remaining locked cosmological constraints. Theorem 3 and the shape-inversion result below are documented in a companion short note submitted in parallel [12].

8.3 Shape-Inversion Obstruction (SIM152)

SIM151 establishes that the self-consistent coupled chameleon coupling of Option B is $\beta_0^* = 1.018$ in the galactic regime with cosmological screening $\beta_\infty \lesssim 2.4 \times 10^{-10}$. SIM152 then tests the resulting monotone profile $\beta(\rho_b) = \beta_0 \tanh^2(\rho_b/\rho_{\text{sc}})$ against the NGC 3198 SPARC rotation curve (43 data points) and finds:

Theorem 4 (Shape-Inversion: Monotone Coupling Cannot Flatten Rotation Curves). *Let $\beta(\rho_b)$ be a monotonically increasing function of baryonic density $\rho_b(r)$ (which decreases with galactic radius r). Then the gravitational enhancement factor $E(r) \equiv V_{\text{obs}}^2(r)/V_{\text{bary}}^2(r)$, as predicted by the coupling, is monotonically decreasing with r . But any flat rotation curve satisfying $V_{\text{obs}}(r) \approx \text{const}$ while $V_{\text{bary}}(r)$ falls in the outer disk requires $E(r)$ monotonically increasing. The model and requirement are anti-correlated; the Pearson correlation between $E_{\text{model}}(r)$ and $E_{\text{required}}(r)$ is $r = -0.81$. No choice of $(\beta_0, \rho_{\text{sc}})$ can reconcile anti-correlated monotone functions.*

The numerical confirmation is stark: the best achievable $\chi^2/\text{dof} = 91.9$ (acceptance threshold ≤ 2.0), with the optimal transition density ρ_{sc} pinned at its lower boundary—effectively the uniform limit where chameleon variation is minimised. The required enhancement values at four reference radii are $E_{\text{req}} = \{1.8, 2.6, 3.8, 5.2\}$ at $r = \{2, 10, 20, 40\}$ kpc; the model produces $E_{\text{model}} = \{3.1, 1.07, 1.04, 1.04\}$ —large where the requirement is small, negligible where the requirement is large.

Theorem 4 applies to any monotonically increasing $\beta(\rho_b)$, not just the tanh profile. An anti-chameleon profile (decreasing with ρ_b) would give E_{model} increasing with r , but would also be large at cosmological densities $\rho_b \sim \rho_{\text{crit}} \ll \rho_{\text{sc}}$, directly violating the SIM151 screening bound $\beta_\infty \lesssim 2.4 \times 10^{-10}$. A non-monotone profile peaking at intermediate galactic densities—large at ρ_{gal} , small at ρ_{inner} and ρ_{cosmo} —is not derivable from the SIM151 ansatz and would constitute a qualitatively different architecture.

8.4 Condensate Mass-Gap Obstruction (SIM153)

Pre-check 1 of SIM153 confirms that the sextic-stabilized condensate (Option C) has the correct qualitative shape: a top-hat soliton produces $E(r)$ increasing at large r (unlike Option B’s structural inversion), with all four reference points within a factor of 2 of the required values. Pre-check 2 then finds three independent structural obstructions that prevent the mechanism from working quantitatively.

Obstruction A (mass gap). The condensate mass required to reproduce the NGC 3198 flat outer curve is $M_C = 7.2 \times 10^{10} M_\odot$. The cosmological dark matter density requires $M_C^{\text{cosmo}} = \rho_{\text{DM}}/n_{\text{gal}} = 3.35 \times 10^{13} M_\odot$ per L^* galaxy ($n_{\text{gal}} = 10^{-3} \text{Mpc}^{-3}$). The mass gap is $M_C^{\text{cosmo}}/M_C = 466$; galactic condensates supply only 0.21% of Ω_{DM} .

Obstruction B (gap-closing breaks curves). Extending the condensate core radius to close the mass gap requires $r_c \gtrsim 232 \text{ kpc}$, larger than the virial radius of an NGC 3198-type halo ($\sim 150 \text{ kpc}$). Inside r_c the density profile is approximately uniform, giving a solid-body velocity profile $V_C(r) \propto r$, which is incompatible with the observed flat $V_{\text{obs}}(r) \approx \text{const}$ across the 5–44 kpc range. The SIM153 numerical test confirms

this: the extended condensate under-predicts V_{obs} by 25–35% at intermediate radii and marginally over-predicts at the outermost point; no choice of r_c simultaneously satisfies both constraints.

Obstruction C (coupling sign). A non-topological soliton (Q-ball or thick-wall condensate) requires at least one attractive term in the scalar potential: negative m_0^2 or negative quartic coupling λ . The locked action has $m_0^2 > 0$ and $\lambda = 0$ (or $\lambda > 0$), both repulsive. Adding $\mu_6 \Psi^6$ with $\mu_6 > 0$ does not produce a soliton minimum without simultaneously flipping the sign of m_0^2 or introducing $\lambda < 0$. Either modification breaks the Phase 1 cosmological fit (SIM113, SIM88), since the entire locked parameter set $(\Lambda_0, \bar{\Psi}, m_0, \lambda)$ is calibrated jointly.

8.5 Synthesis: Three Channels Exhausted

Theorem 5 (Structural Irreducibility of Galactic Dark Matter in CMSTG). *Within the locked Phase 1 CMSTG action, the three available channels of action extension—mass modification (Option A), coupling modification (Option B), and potential modification (Option C)—each fail to produce galactic flat rotation curves by an independent structural obstruction:*

1. Mass modification (SIM112, SIM149–SIM150): *the G_{eff}/G ceiling (Theorem 3) is $\sim 3\times$ below the galactic requirement at any Solar-System-consistent Λ_0 .*
2. Coupling modification (SIM151–SIM152): *any monotone $\beta(\rho_b)$ coupling produces a shape inversion anti-correlated with the flat-curve requirement (Theorem 4); non-monotone profiles require architecture incompatible with cosmological screening.*
3. Potential modification (SIM153): *the sextic-stabilized condensate faces a $466\times$ mass gap between galactic-fit and cosmological dark matter density, with all gap-closing routes blocked by rotation-curve shape incompatibility and coupling-sign constraints.*

Combining Sections 6 and 8: every mechanism and action-extension channel within the CMSTG action class has been exhausted. Dark matter does not arise from within the locked Phase 1 CMSTG action through any of the three available channels.

This result is the galactic analogue of the dark-energy no-go of Paper I: as Paper I proves the DESI residual is structurally irreducible within the action class, this paper and the Phase 5 results establish that dark matter is structurally absent from the action class. The implication for cosmology is that if CMSTG is the correct theory of gravity, dark matter in the observed universe must come from a sector structurally distinct from the scalar-tensor framework: standard cold dark matter, axions, sterile neutrinos, and other candidates remain fully compatible with CMSTG’s cosmological successes (Papers I–III); they are simply not generated by it.

9 Conclusions

We have established that the Phase 1 canonical locked action of CMSTG cannot replace cold dark matter at galactic scales through any mechanism or action-extension channel accessible within the action class. Simulations SIM99–SIM103 test the mechanisms present in the locked action itself; SIM149–SIM153 complete the test of the three proposed extension options (Options A, B, C). The results are:

1. **Static profile (SIM99):** The Yukawa field profile is exponentially steeper than the isothermal halo requirement. The quartic potential modifies the central amplitude but not the exponential radial dependence.
2. **Time-domain dynamics (SIM100):** The curvature coupling $2\Lambda_0|R_{\text{gal}}|$ is $\sim 2 \times 10^6$ times weaker than m_0^2 at all galactic densities. Condensation requires neutron-star densities. No quartic self-coupling produces a flat-curve condensate: the sub-threshold window gives Yukawa profiles; the super-threshold window gives runaway blow-up.
3. **Vainshtein screening (SIM103):** The Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full (α_V, M_V) parameter space. The field gradient is 10^4 – $10^6 \times$ below the nonlinear threshold at all galactic radii.
4. **Memory-kernel G_{eff} boost (SIM149–SIM150):** The theoretical ceiling $G_{\text{eff}}/G_N = 1 + 2\Lambda_0\Psi_0^2 = 1.041$, achieved when $\Psi_{\text{local}} \rightarrow 0$, falls a factor $\sim 3 \times$ short of the 3.115 required by NGC 3198. A Λ_0 sweep (SIM150) confirms the ceiling is universal: the only Λ_0 that lifts it to 3.115 ($\Lambda_{0,\text{req}} = 0.154 M_{\text{Pl}}^{-2}$) violates the Cassini PPN bound by $\sim 2 \times 10^4 \times$ and all five remaining locked constraints (Theorem 3).
5. **Chameleon coupling shape inversion (SIM151–SIM152):** The self-consistent coupled chameleon coupling $\beta_0^* = 1.018$ (SIM151) gives a density-dependent profile that is monotonically decreasing with galactic radius, while flat rotation curves require the enhancement factor $E(r)$ to be monotonically increasing. The Pearson anti-correlation is $r = -0.81$; best achievable $\chi^2/\text{dof} = 91.9$ (Theorem 4).
6. **Sextic-condensate mass gap (SIM153):** A sextic-stabilized condensate has the correct qualitative rotation-curve shape but faces a $466 \times$ mass gap between the galactic-fit mass ($7.2 \times 10^{10} M_\odot$) and the cosmological dark matter requirement ($3.35 \times 10^{13} M_\odot$). Closing the gap by extending the condensate radius produces a solid-body velocity profile; closing it by a coupling-sign flip breaks the Phase 1 cosmological fit (Theorem 5).

The energy density deficit at $r = 10 \text{ kpc}$ is $2.7 \times 10^6 \times$ under all mechanisms within the locked action. Phase 5 (Section 8) completes the exhaustion of the three extension options. This is a clean negative result: every mechanism and action-extension channel within the CMSTG action class has been tested and every mechanism fails by a structural argument, not a parameter choice. Dark matter in the observed universe must therefore come from a sector structurally distinct from the CMSTG framework—standard cold dark matter, axions, sterile neutrinos, and other well-motivated candidates remain fully compatible with CMSTG’s cosmological successes; they are simply not generated by it.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity (simulations SIM99, SIM100, SIM103, SIM147–SIM153).

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